

CSCI 6850
Project 2
Atomic Charge Prediction with Graph Neural Networks (GNNs)

Background

Atomic charge (AC), which represents the charge distribution within a molecule, is an important property influencing molecular structures, reactivity, and inter/intra-molecular interactions. Machine learning approaches, particularly Graph Neural Networks (GNNs), have been increasingly used for atomic charge prediction by learning from molecular structures. These models provide efficient and accurate predictions compared to traditional quantum mechanical calculations.

In this project, you will develop a GNN model to predict atomic charges from molecular structures using **POSCAR files** as input.

Molecular Representation

A **POSCAR file** is a plain text file used in computational materials science to define atomic structures, including:

1. **Comment Line** – A descriptive title for the system.
2. **Scaling Factor** – A global scaling factor.
3. **Lattice Vectors** – A 3×3 matrix defining the unit cell.
4. **Element Symbols** (*optional*) – Lists atomic species.
5. **Number of Atoms** – Specifies the count of each atomic species.
6. **Selective Dynamics** (*optional*) – Indicates whether atoms can move.
7. **Coordinate Type** – Either **Direct (fractional)** or **Cartesian**.
8. **Atomic Positions** – Lists atomic coordinates.

Here is an example **POSCAR** file containing two types of elements, **Silicon (Si)** and **Oxygen (O)**, in a simple structure:

```
SiO2 - Quartz Structure
1.0 # Scaling factor
4.913 0.000 0.000 # Lattice vector a
0.000 4.913 0.000 # Lattice vector b
0.000 0.000 5.405 # Lattice vector c
Si O # Element symbols
2 4 # Number of atoms: 2 Si, 4 O
Direct # Direct (fractional) coordinates
0.4697 0.0000 0.6667 # Si atom 1
0.5303 0.5000 0.3333 # Si atom 2
0.4133 0.2670 0.1188 # O atom 1
0.4133 0.7330 0.8812 # O atom 2
0.5867 0.7330 0.1188 # O atom 3
0.5867 0.2670 0.8812 # O atom 4
```

Explanation:

- **SiO2 - Quartz Structure** → A descriptive comment.
- **1.0** → Global scaling factor.
- **Lattice vectors** → Defines unit cell dimensions.
- **Si O** → Two element types.
- **2 4** → Number of Si and O atoms.
- **Direct** → Specifies fractional coordinates.
- **Atomic positions** → Each row represents an atom's coordinates.

Periodic Boundary Condition (PBC) in the POSCAR File

In computational chemistry, **Periodic Boundary Conditions (PBC)** are used to simulate infinite crystalline structures with a finite unit cell:

- The unit cell **repeats infinitely** in all directions.
- Atoms near one boundary interact with atoms near the opposite boundary.
- This eliminates artificial surface effects and ensures realistic simulations.

Using ASE to Compute Pairwise Distances

ASE is a powerful Python library for working with atomic structures and simulations.

Key Features:

- Reads/writes structure files (POSCAR, CIF, XYZ).
- Builds/manipulates atomic structures.
- Computes distances, angles, and bonds.
- Interfaces with electronic structure codes like VASP and Quantum ESPRESSO.

Example: Compute Pairwise Distances from a POSCAR File

```
from ase.io import read

atoms = read("POSCAR") # Read the POSCAR file
distances = atoms.get_all_distances(mic=True) # Compute distances with PBC
print(distances)
```

- `read("POSCAR")`: Loads the atomic structure.
- `atoms.get_all_distances(mic=True)`: Computes pairwise distances while automatically applying **PBC** using the Minimum Image Convention.
- The output is a **distance matrix**, where `distances[i, j]` gives the distance between atom *i* and atom *j*.

Dataset for this project

The dataset consists of **two types of files**:

1. **POSCAR files** – Define atomic structures.
2. **Atomic charge files** – Contain atomic charges corresponding to each atom in the POSCAR files.

The data files are uploaded to the **shared** folder (/ocean/projects/cis250151p/shared/Data). You should have access to it.

Note: please do not duplicate these files into your own folder to save disk space.

Project Tasks

1. Code Implementation

- Create a Graph Dataset
 - Convert the dataset into a graph format suitable for a Graph Neural Network (GNN).
 - Use PyTorch Geometric to create the dataset.
 - Refer to

[Introduction by Example — pytorch_geometric documentation](#)

[Creating Graph Datasets — pytorch_geometric documentation](#)

Bonus Opportunity (10 Points):

If you write your own code to create the **PyG dataset** from the provided data files (strongly encouraged!), you will receive **10 bonus points**. Otherwise, I will provide the PyG dataset file.

- **Develop a GNN Model**
 - Train a GNN to **predict atomic charges** based on molecular structures.
 - Split the dataset into **training, validation, and test sets**.
 - Tune **hyperparameters** to maximize accuracy.

2. Technical Report (8+ Pages)

Your report should include:

- **Introduction:** Explain the problem, motivation, and background.
- **Dataset Description:** Describe the POSCAR and atomic charge data.
- **Methodology:** Explain how the graph dataset is created, the GNN model design, and the training procedure (including hyperparameter tuning).
- **Results:** Present prediction accuracy, loss curves, and evaluation metrics.
Note: you should avoid overfitting and under-fitting.
- **Discussion:** Analyze results, challenges faced, and potential improvements.
- **Conclusion:** Summarize findings and future work.

3. Deliverables

1. **Code:** Submit a well-documented Python script.
2. **Technical Report:** Submit a PDF with all required sections.

Happy coding and good luck!